

Joseph Agossa

Office Contact Information

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Research Interests

Econometrics, Big Data, Machine Learning / Text as Data (NLP), Large Language Models (LLMs), Generative AI.

ESG Disclosure, Corporate Governance, Development Economics, Health Economics, Applied Statistics.

Education

2019–2020: MSc in Statistics and Econometrics (International Track), Toulouse School of Economics (TSE), France.

2018–2019: MSc 1 in Applied Mathematics for Engineering, Industry and Innovation (MAPI3), Université Toulouse 3 Paul Sabatier, France.

2015–2016: BSc in Mathematics and Computer Science Applied to Human and Social Sciences (MIASHS), Université Toulouse 2 Jean Jaurès, France.

2012–2013: DEUG in Applied Mathematics, Ibn Tofail University, Morocco.

Publications

1. “CADA-PRO: A Patient Questionnaire Measuring Key Cognitive, Motor, Emotional, and Behavioral Outcomes in CADASIL”, with C. Di Folco, A. Jabouley, S. Reyes, C. Machado, S. Guey, D. Hervé, F. Fernandes, H. Chabriat, and S. Tezenas du Montcel, *Stroke*, Vol. 55, pp. 2439–2448, 2024.
2. “Quantitative Analysis of Cell Aggregation Dynamics Identifies HDAC Inhibitors as Potential Regulators of Cancer Cell Clustering”, with F. Gava, J. Pignolet, S. Déjean, O. Mondésert, R. Morin, B. Ducommun, and V. Lobjois, *Cancers*, Vol. 13, 5840, 2021.

Research and Work Experience

Aug 2022 – present: Research Engineer – Data Scientist & AI Ambassador, ESSEC Business School, Research Center (CERESSEC), Cergy, France.

Jun 2021 – Jul 2022: Data Scientist, INSERM – IPLESP & Brain and Spine Institute (ICM), Paris, France.

May 2020 – Nov 2020: Biostatistics–Epidemiology Intern (Real World Solutions), IQVIA Operations France, La Défense.

Oct 2019 – Apr 2020: Statistical Consultant (part-time), BVA Group, Toulouse, France.

May 2019 – Sep 2019: Master 1 Intern in Biostatistics, CNRS, Toulouse, France.

Jul 2017 – Jan 2018: Statistics–Computer Science Intern, IHSES France, Toulouse.

Teaching

Instructor, ESSEC Business School & CentraleSupélec

2024 – present

- Python for Data Analysis, Global BBA / EPSCI (1st cycle), ESSEC.
- Advanced Python for Data Analysis (elective), Global BBA (2nd cycle), ESSEC.
- Big Data Algorithms, Techniques & Platforms, Master DSBA (ESSEC & CentraleSupélec): MapReduce, Spark, Spark DataFrames, MongoDB.

Certifications

- Data Science: Machine Learning — Harvard University (2022).
- SAS Certified Specialist: Base Programming Using SAS 9.4 (2021).
- Big Data, Hadoop and Spark Basics — IBM.
- SQL for Data Science — IBM.
- Compilation Basics for Macroeconomic Statistics (2021).

Skills and Languages

Programming: Python, R, SAS, STATA, SPSS, SQL, C/C++, Fortran, JavaScript, PHP, HTML, MATLAB.

Machine Learning: Regression (linear/logistic), Random Forest, Decision Trees, K-means, Clustering, Cross-Validation.

Deep Learning: TensorFlow, Keras, PyTorch, DNN, RNN, LSTM, CNN, Text mining.

Methods: Econometrics, Time series, Survey sampling, Spatial econometrics, Scoring, Non-parametric models.

Big Data: Hadoop, Spark, MapReduce, MongoDB.

Languages: French (native), English (fluent).